

Signorini's Problem

Max Stuthmann

March 19, 2026

- 1 Introduction – What's the Problem?
- 2 Geometrical and Physical Setup
 - Boundary Decomposition
 - Elasticity and Forces
 - Admissible Displacements
- 3 Energy Minimization
- 4 Weak Formulations
- 5 Strong Formulations
 - A Hybrid Formulation
 - Classical Strong Formulation
- 6 Mixed Formulation

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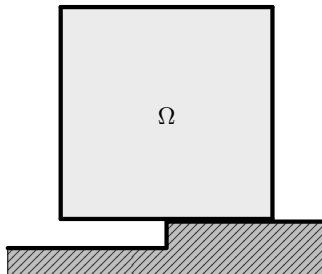
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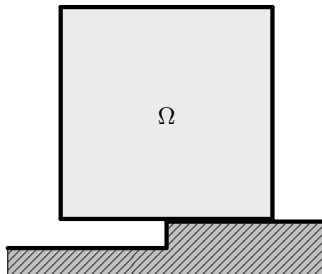
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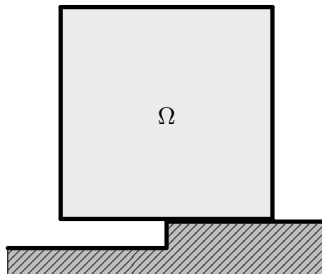
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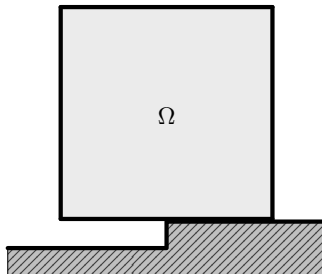
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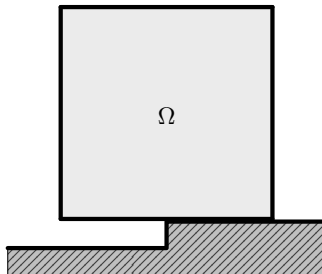
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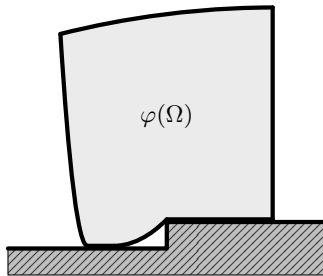
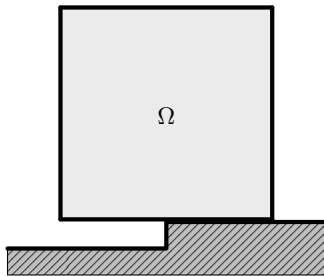
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- ▶ **Not in equilibrium configuration.**



What is to be found?

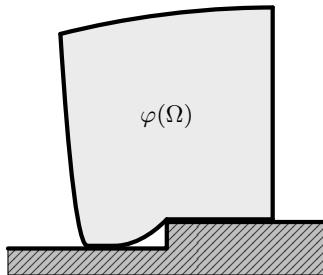
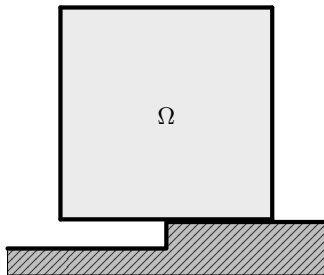
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$$\varphi(x) = x + u(x), \quad u : \bar{\Omega} \rightarrow \mathbb{R}^d.$$

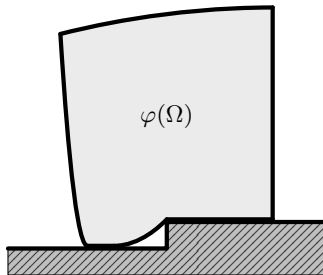
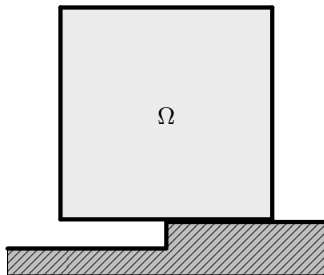


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- ▶ We aim to find the *displacement function* u .



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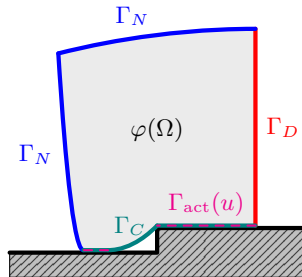
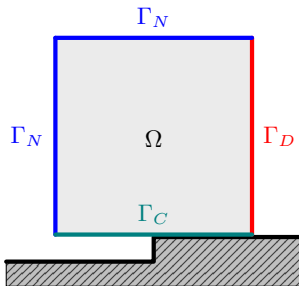
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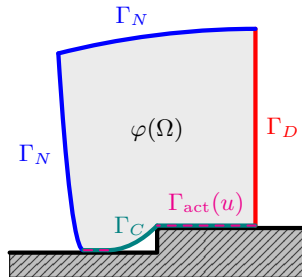
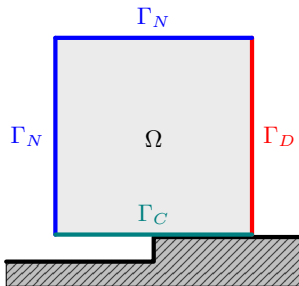
Boundary Decomposition and Preview on the Boundary Conditions

- Decompose $\Gamma = \partial\Omega$ into three relatively open and pairwise disjoint parts $\Gamma = \overline{\Gamma_D} \dot{\cup} \overline{\Gamma_N} \dot{\cup} \overline{\Gamma_C}$ with $|\Gamma_D|, |\Gamma_C| > 0$ and $|\Gamma_N| \geq 0$.



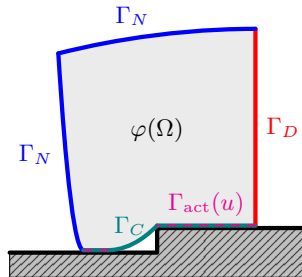
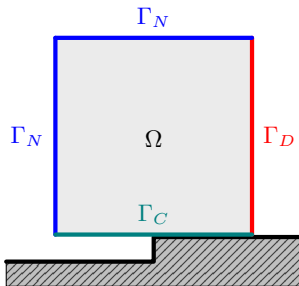
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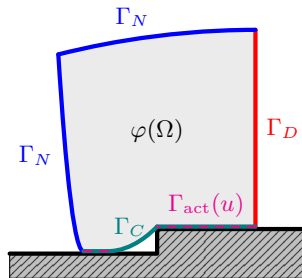
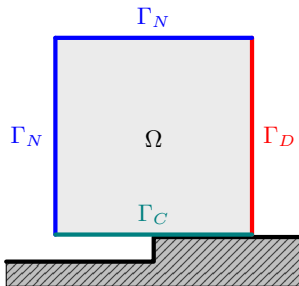
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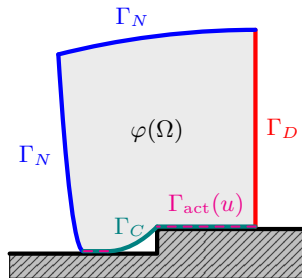
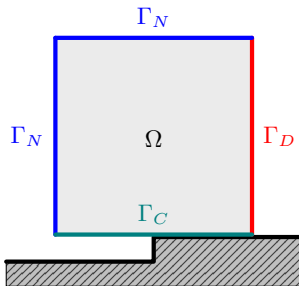
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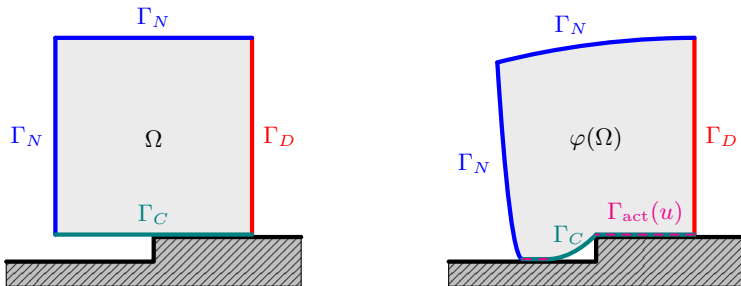
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- ▶ Rigid foundation can be describe by gap-function $g : \Gamma_C \rightarrow \mathbb{R}$. We assume $g = 0$.



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Basic Notations on Sobolev Spaces

- ▶ On $H^s(\Omega; \mathbb{R}^d)$ for $s = k + \theta$ with $k \in \mathbb{N}_0$ and $\theta \in [0, 1)$, we use $(\cdot, \cdot)_{s, \Omega}$ and $\|\cdot\|_{s, \Omega}$

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- ▶ *Trace Theorem*: Restriction of smooth functions $\overline{\Omega} \rightarrow \mathbb{R}^d$ onto the boundary $\Gamma = \partial\Omega \in \mathcal{C}^{m,1}$ extends uniquely to a continuous surjective linear trace operator

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- ▶ For now, Ω is Lipschitz, i.e. $m = 0$ and we need $s = 1$, so

$$\gamma : H^1(\Omega; \mathbb{R}^d) \rightarrow H^{1/2}(\Gamma; \mathbb{R}^d)$$

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- *Symmetric second order tensors:*

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- If $\mathcal{A} : \Omega \rightarrow \mathcal{L}(\mathbb{S}^d; \mathbb{S}^d)$ is a fourth order tensor and $s \in \mathbb{S}^d$, we use the *double contraction*

$$s : \mathcal{A}(x) : s := \sum_{i=1}^d \sum_{j=1}^d \sum_{k=1}^d \sum_{l=1}^d \mathcal{A}_{ijkl}(x) s_{ij} s_{kl}.$$

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- ▶ $\mathcal{A}_{ijkl} \in L^\infty(\Omega)$ is sufficient for $s = 1$. Later on, we need more.

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- ▶ *Work = force · displacement:* The linear functional

$$\ell(v) := \int_{\Omega} f \cdot v \, dx + \int_{\Gamma_N} F \cdot (\gamma v) \, ds, \quad v \in H^1(\Omega; \mathbb{R}^d)$$

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- ▶ *Equilibrium* between both: $a(u, v) = \ell(v) \quad \forall v \in H^1(\Omega; \mathbb{R}^d)$.

Standard Properties of a and ℓ

Proposition

The symmetric bilinear form $a : H^1(\Omega; \mathbb{R}^d) \times H^1(\Omega; \mathbb{R}^d) \rightarrow \mathbb{R}$ defined above is

i) bounded, i.e. for all $v, w \in H^1(\Omega; \mathbb{R}^d)$ there exists $C_1 < \infty$ such that

$$|a(u, v)| \leq C_1 \|u\|_{1,\Omega} \|v\|_{1,\Omega}.$$

ii) $H^1(\Omega; \mathbb{R}^d)$ -elliptic, i.e. for all $v \in H^1(\Omega; \mathbb{R}^d) \setminus \{0\}$ there exist $C_2 > 0$ such that

$$a(v, v) \geq C_2 \|v\|_{1,\Omega}^2.$$

Moreover, the linear functional ℓ defined above is bounded, i.e. for all $v \in H^1(\Omega; \mathbb{R}^d)$ there exists $C_3 > 0$ such that

$$|\ell(v)| \leq C_3 \|v\|_{1,\Omega}.$$

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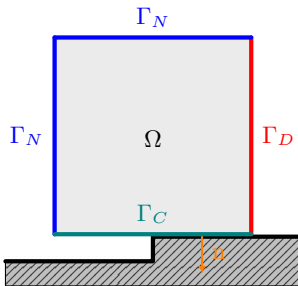
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Admissible Displacements and the Nonpenetration Constraint

- *Homogeneous Dirichlet* boundary conditions on Γ_D are incorporated in the underlying Hilbert space

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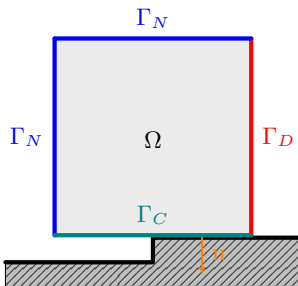


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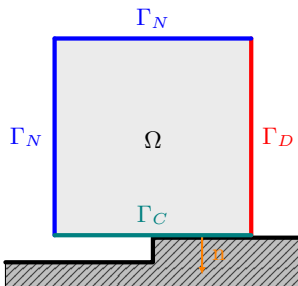


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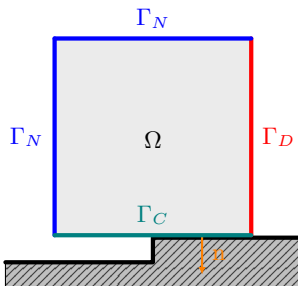


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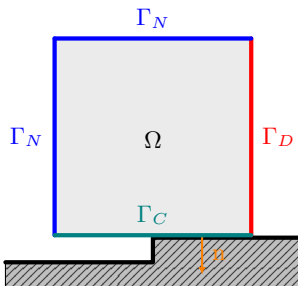


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- Boundary Decomposition
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► *Equilibrium:*

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Problem (Signorini's Problem – Energy Minimization)

$$\text{Find } u \in K \text{ such that: } \mathcal{J}(u) = \min_{v \in K} \mathcal{J}(v).$$

Proposition (Well-Posedness of Signorini's Problem)

There exists exactly one $u \in K$ that minimizes $\mathcal{J}$ over K . Moreover, the unique minimizer satisfies the a-priori bound

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- Admissible Displacements

3 Energy Minimization

4 Weak Formulations

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Proposition (First-order optimality condition on convex sets)

A function $u \in K$ minimizes $\mathcal{J}$ over K if and only if it satisfies the variational inequality

$$\mathcal{J}'(u)[v - u] = a(u, v - u) - \ell(v - u) \geq 0 \quad \forall v \in K.$$

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- Boundary Decomposition
- Elasticity and Forces
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2 Geometrical and Physical Setup

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- ▶ Then $\operatorname{div} \sigma(u) \in L^2(\Omega; \mathbb{R}^d)$ and we can understand normal traces on $H(\operatorname{div}; \Omega)$ in the following way.

Normal Traces on $H(\operatorname{div}; \Omega)$

- ▶ We denote by $H^{-1/2}(\Gamma; \mathbb{R}^d)$ the dual space of $H^{1/2}(\Gamma; \mathbb{R}^d)$ with duality pairing $\langle \cdot, \cdot \rangle_\Gamma$.

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- ▶ In the smooth case, $\gamma_n(\tau)$ coincides with the classical traction $(\gamma\tau)_n$.

Stress Regularity Assumption

- ▶ Thus, we further assume $\sigma(u) \in H(\operatorname{div}; \Omega)$ which yields $\sigma(u)_n := \gamma_n(\sigma(u)) \in H^{-1/2}(\Gamma; \mathbb{R}^d)$.

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$$\langle \sigma(u)n, \gamma v \rangle_\Gamma = \int_\Omega \sigma(u) : \varepsilon(v) \, dx + \int_\Omega \operatorname{div} \sigma(u) \cdot v \, dx \quad \forall v \in V.$$

Stress Regularity Assumption

- ▶ Thus, we further assume $\sigma(u) \in H(\operatorname{div}; \Omega)$ which yields $\sigma(u)n := \gamma_n(\sigma(u)) \in H^{-1/2}(\Gamma; \mathbb{R}^d)$.
- ▶ Since $\sigma(u)$ is symmetric, we have

$$\sigma(u) : \nabla v = \sigma(u) : \varepsilon(v) \quad \forall v \in V.$$

- ▶ Therefore

$$\langle \sigma(u)n, \gamma v \rangle_\Gamma = \int_\Omega \sigma(u) : \varepsilon(v) \, dx + \int_\Omega \operatorname{div} \sigma(u) \cdot v \, dx \quad \forall v \in V.$$

- ▶ *Useful identity:* If $-\operatorname{div} \sigma(u) = f$ almost everywhere in Ω , then

$$a(u, v) - \ell(v) = \langle \sigma(u)n, \gamma v \rangle_\Gamma - \int_{\Gamma_N} F \cdot (\gamma v) \, ds, \quad \forall v \in V.$$

A Hybrid Formulation

Problem (Signorini's Problem – Hybrid Formulation)

For given $f \in L^2(\Omega; \mathbb{R}^d)$ and $F \in L^2(\Gamma_N; \mathbb{R}^d)$, find $u \in K$ such that

$$\begin{cases} -\operatorname{div} \sigma(u) = f & \text{a.e. in } \Omega, \end{cases} \quad (1a)$$

$$\begin{cases} \langle \sigma(u)n, \gamma u \rangle_\Gamma = \int_{\Gamma_N} F \cdot (\gamma u) \, ds, \end{cases} \quad (1b)$$

$$\begin{cases} \langle \sigma(u)n, \gamma v \rangle_\Gamma \geq \int_{\Gamma_N} F \cdot (\gamma v) \, ds \quad \forall v \in K. \end{cases} \quad (1c)$$

Equivalence of Hybrid and Weak Formulation

Proposition (Equivalence of Hybrid and Weak Formulation)

Assume $\sigma(u) \in H(\operatorname{div}; \Omega)$. Then $u \in K$ solves the hybrid problem above if and only if it solves the weak problem

$$\text{Find } u \in K \text{ such that} \quad a(u, v - u) \geq \ell(v - u) \quad \forall v \in K.$$

Proof: *Hybrid $\Rightarrow$ Weak:*

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- ▶ Subtract (1b) from (1c):

$$\langle \sigma(u)\mathbf{n}, \gamma(v - u) \rangle_{\Gamma} - \int_{\Gamma_N} F \cdot \gamma(v - u) \, ds \geq 0 \quad \forall v \in K.$$

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- ▶ Insert the identity from the above:

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i.e. the weak formulation.

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► Let $\varphi \in \mathcal{D}(\Omega; \mathbb{R}^d)$. Then $u \pm t\varphi \in K$ for $t \in \mathbb{R}$ because $\gamma\varphi = 0$, hence the weak inequality implies

$$a(u, \varphi) = \ell(\varphi) = \int_{\Omega} f \cdot \varphi \, dx = \langle f, \varphi \rangle \quad \forall \varphi \in \mathcal{D}(\Omega; \mathbb{R}^d).$$

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- ▶ Thus, $-\operatorname{div} \sigma(u) = f$ in $\mathcal{D}'(\Omega; \mathbb{R}^d)$ and by $\sigma(u) \in H(\operatorname{div}; \Omega)$, it holds in $L^2(\Omega; \mathbb{R}^d)$ as well.

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- ▶ This is exactly (1b).

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- ▶ This means

$$\langle \sigma(u)n, \gamma v \rangle_{\Gamma} \geq \int_{\Gamma_N} F \cdot \gamma v \, ds + \underbrace{\left[\langle \sigma(u)n, \gamma u \rangle_{\Gamma} - \int_{\Gamma_N} F \cdot \gamma u \, ds \right]}_{=0} \quad \forall v \in V.$$

1 Introduction – What's the Problem?

2 Geometrical and Physical Setup

- Boundary Decomposition
- Elasticity and Forces
- Admissible Displacements

3 Energy Minimization

4 Weak Formulations

5 Strong Formulations

- A Hybrid Formulation
- Classical Strong Formulation

6 Mixed Formulation

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- ▶ Normal Trace: $t(u) := (\gamma \sigma(u))_n \in H^{s-3/2}(\Gamma; \mathbb{R}^d)$ is well-defined.
- ▶ If $\sigma(u) \in H(\text{div}; \Omega)$, the two concepts of normal traces coincide:

$$\langle \sigma(u)_n, \gamma v \rangle_\Gamma = \int_\Gamma t(u) \cdot (\gamma v) \, ds \quad \forall v \in V.$$

Normal and Tangential Components

- Normal and tangential splitting of $t(u)$ into

$$\sigma_n(u) := t(u) \cdot \mathbf{n} \in H^{s-3/2}(\Gamma), \quad \sigma_t(u) := t(u) - \sigma_n(u) \mathbf{n} \in H^{s-3/2}(\Gamma; \mathbb{R}^d).$$

such that

$$t(u) = \sigma_n(u) \mathbf{n} + \sigma_t(u).$$

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- Furthermore, for $u \in H^s(\Omega; \mathbb{R}^d)$ we have $\gamma u \in H^{s-1/2}(\Gamma; \mathbb{R}^d)$ with normal/tangential splitting

$$u_n := (\gamma u) \cdot n \in H^{s-1/2}(\Gamma) \quad \text{and} \quad u_t := \gamma u - u_n n \in H^{s-1/2}(\Gamma; \mathbb{R}^d)$$

such that

$$\gamma u = u_n n + u_t.$$

Classical Strong Formulation under Additional Regularity

Problem (Signorini's Problem – Classical Strong Formulation)

For given $f \in L^2(\Omega; \mathbb{R}^d)$ and $F \in L^2(\Gamma_N; \mathbb{R}^d)$ find $u \in H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$ such that the small strain elasticity equations

$$\begin{cases} -\operatorname{div} \sigma(u) = f & \text{in } \Omega, \\ u = 0 & \text{on } \Gamma_D, \\ t(u) = F & \text{on } \Gamma_N, \end{cases} \quad \begin{array}{l} (2a) \\ (2b) \\ (2c) \end{array}$$

and the contact conditions

$$\begin{cases} u_n \leq 0 & \text{on } \Gamma_C, \\ \sigma_n(u) \leq 0 & \text{on } \Gamma_C, \\ \sigma_n(u) u_n = 0 & \text{on } \Gamma_C, \\ \sigma_t(u) = 0 & \text{on } \Gamma_C \end{cases} \quad \begin{array}{l} (3a) \\ (3b) \\ (3c) \\ (3d) \end{array}$$

hold.

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- ▶ *Complementarity* $\sigma_n(u) u_n = 0$ on Γ_C : either there is no contact pressure, or the body is in contact, but not both effects independently.

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- ▶ *Compressive contact stress* $\sigma_n(u) \leq 0$ on Γ_C : the foundation can exert only compressive normal forces.
- ▶ *Complementarity* $\sigma_n(u) u_n = 0$ on Γ_C : either there is no contact pressure, or the body is in contact, but not both effects independently.
- ▶ *Frictionless contact* $\sigma_t(u) = 0$ on Γ_C : Foundation does not exert any tangential force on the body.

Proposition (Equivalence of Strong and Hybrid Formulation)

Let $u \in H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$. Then u solves the strong problem if and only if it solves the hybrid problem.

Proof: *Strong $\implies$ Hybrid:*

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Proof: Strong $\implies$ Hybrid:

Problem (Signorini's Problem – Hybrid Formulation)

For given $f \in L^2(\Omega; \mathbb{R}^d)$ and $F \in L^2(\Gamma_N; \mathbb{R}^d)$, find $u \in K$ such that

$$\begin{cases} -\operatorname{div} \sigma(u) = f & \text{a.e. in } \Omega, \end{cases} \quad (4a)$$

$$\begin{cases} \langle \sigma(u)n, \gamma u \rangle_\Gamma = \int_{\Gamma_N} F \cdot (\gamma u) \, ds, \end{cases} \quad (4b)$$

$$\begin{cases} \langle \sigma(u)n, \gamma v \rangle_\Gamma \geq \int_{\Gamma_N} F \cdot (\gamma v) \, ds \quad \forall v \in K. \end{cases} \quad (4c)$$

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Proof: *Strong $\implies$ Hybrid (Bulk equation and $u \in K$):*

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- ▶ Let $u \in H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$ solve the strong problem.
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- ▶ Let $u \in H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$ solve the strong problem.
- ▶ Bulk Equation $-\operatorname{div} \sigma(u) = f$ appears in both.
- ▶ $u = 0$ on Γ_D (2b) and $u_n \leq 0$ on Γ_C (3a) are incorporated in the requirement $u \in K$.

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Let $u \in H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$. Then u solves the strong problem if and only if it solves the hybrid problem.

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► We want to show

$$\langle \sigma(u)n, \gamma v \rangle_\Gamma = \int_\Gamma t(u) \cdot (\gamma v) \, ds \geq \int_{\Gamma_N} F \cdot (\gamma v) \, ds \quad \forall v \in K.$$

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► With $t(u) = \sigma_n(u)n + \sigma_t(u)$ and $\gamma v = v_n n + v_t$, we have

$$t(u) \cdot (\gamma v) = \sigma_n(u) v_n + \underbrace{\sigma_t(u) \cdot v_t}_{=0 \text{ on } \Gamma_C \text{ since } \sigma_t(u)=0 \text{ on } \Gamma_C}.$$

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- Since $v \in K$, we have $v_n \leq 0$ on Γ_C and by (3b), we have $\sigma_n(u) \leq 0$ on Γ_C .
- Thus $\sigma_n(u) v_n \geq 0$ on Γ_C and thus

$$\langle \sigma(u)n, \gamma v \rangle_\Gamma \geq \int_{\Gamma_N} F \cdot (\gamma v) \, ds \quad \forall v \in V.$$

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- ▶ Let $u \in H^s(\Omega; \mathbb{R}^d)$, $s > 3/2$ solve the hybrid problem.
- ▶ Bulk equation $-\operatorname{div} \sigma(u) = f$ is contained in both problems.
- ▶ Since $u \in K$, we have $u = 0$ on Γ_D and $u_n \leq 0$ on Γ_C .
- ▶ It remains to prove $t(u) = F$ on Γ_N and the remaining contact conditions.

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Let $u \in H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$. Then u solves the strong problem if and only if it solves the hybrid problem.

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- ▶ Then $\pm v \in K$.
- ▶ Plug $\pm v$ into hybrid inequality:

$$\int_{\Gamma_N} t(u) \cdot g \, ds \geq \int_{\Gamma_N} F \cdot g \, ds, \quad - \int_{\Gamma_N} t(u) \cdot g \, ds \geq - \int_{\Gamma_N} F \cdot g \, ds.$$

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- ▶ Therefore,

$$\int_{\Gamma_N} (t(u) - F) \cdot g \, ds = 0 \quad \forall g \in H_{00}^{1/2}(\Gamma_N; \mathbb{R}^d) \quad \implies \quad t(u) = F \text{ on } \Gamma_N.$$

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- ▶ Let $g_t \in H_{00}^{1/2}(\Gamma_C; \mathbb{R}^d)$ with $g_t \cdot \mathbf{n} = 0$, and choose $v \in H^1(\Omega; \mathbb{R}^d)$ with $(\gamma v)|_{\Gamma_C} = g_t$ and $(\gamma v)|_{\Gamma_N} = 0$ and $(\gamma v)|_{\Gamma_D} = 0$.

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- ▶ With $t(u) = \sigma_n(u)\mathbf{n} + \sigma_t(u)$ this means

$$\int_{\Gamma_C} (\sigma_n(u)\mathbf{n} + \sigma_t(u)) \cdot g_t \, ds = \int_{\Gamma_C} \sigma_n(u) \underbrace{(\mathbf{n} \cdot g_t)}_{=0} \, ds + \int_{\Gamma_C} \sigma_t(u) \cdot g_t \, ds = 0.$$

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- ▶ Let $\eta \in H_{00}^{1/2}(\Gamma_C)$ with $\eta \leq 0$ on Γ_C and choose $v \in H^1(\Omega; \mathbb{R}^d)$ with $(\gamma v)|_{\Gamma_C} = \eta n$ and $(\gamma v)|_{\Gamma_N} = 0$ and $(\gamma v)|_{\Gamma_D} = 0$.

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- ▶ With $t(u) = \sigma_n(u) \mathbf{n} + \sigma_t(u)$ and $\sigma_t(u) = 0$ on Γ_C , we have

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- ▶ With $t(u) = \sigma_n(u) \mathbf{n} + \sigma_t(u)$ and $\sigma_t(u) = 0$ on Γ_C , we have

$$\int_{\Gamma_C} t(u) \cdot \gamma u \, ds = \int_{\Gamma_C} \underbrace{\sigma_n(u)}_{\leq 0} \underbrace{u_n}_{\leq 0} \, ds = 0.$$

□

1 Introduction – What's the Problem?

2 Geometrical and Physical Setup

- Boundary Decomposition
- Elasticity and Forces
- Admissible Displacements

3 Energy Minimization

4 Weak Formulations

5 Strong Formulations

- A Hybrid Formulation
- Classical Strong Formulation

6 Mixed Formulation

Preparing for Mixed Formulation

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- ▶ Define contact normal and tangential trace spaces

$$W_{C,n} := \{w \cdot \mathbf{n} \mid w \in W_C\}, \quad W_{C,t} := \{w \in W_C \mid w \cdot \mathbf{n} = 0\}.$$

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- ▶ Define contact normal and tangential trace spaces

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- ▶ Set $\sigma_C(u)\mathbf{n} := \gamma_{n,C}(\sigma(u)) \in W'_C$ with splitting

$$\langle \sigma_C(u)\mathbf{n}, w \rangle_{\Gamma_C} = \langle \sigma_{C,n}(u), w_n \rangle_{\Gamma_C} + \langle \sigma_{C,t}(u), w_t \rangle_{\Gamma_C}, \quad \forall w = w_n \mathbf{n} + w_t \in W_C.$$

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Problem (Signorini's Problem – Mixed Weak Formulation)

For given $f \in L^2(\Omega; \mathbb{R}^d)$ find $(u, \lambda) \in V \times \Lambda_C$ such that

$$\begin{cases} a(u, v) - b(v, \lambda) = \ell(v) & \forall v \in V, \end{cases} \quad (5a)$$

$$\begin{cases} b(u, \mu - \lambda) \geq 0 & \forall \mu \in \Lambda_C. \end{cases} \quad (5b)$$

Equivalence of Weak and Mixed Formulation

Proposition (Equivalence of Weak and Mixed Formulation)

Assume $|\Gamma_N| = 0$. If u solves the weak problem, then the pair $(u, \sigma_{C,n}(u))$ solves the mixed problem. Conversely, if (u, λ) solves the mixed problem, then u solves the weak problem and $\lambda = \sigma_{C,n}(u)$.